

GEBZE TECHNICAL UNIVERSITY
ELECTRONIC ENGINEERING
ELC 337

High Gain Signal Amplification with PNP Active Load Amplifier

Name	Fouad	Büşra
Surname	Aladhami	Ülker
Student ID	210102002808	200102002011

CONTENTS

1 PROCEDURE 3

2 DC ANALYSIS 4

2.1 THEORETICAL 4
2.2 EXPERIMENT 4

3 FREQUENCY RESPONSE OF AMPLIFIER (GAIN, BW) 4

3.1 THEORETICAL 4
3.2 EXPERIMENT 4

4 THEORETICAL TRANSIENT RESPONSE OF AMPLIFIER (IN SIMULATION) 5

5 POWER CONSUMPTION 6

5.1 THEORETICAL 6
5.2 EXPERIMENT 6

6 RIN 7

6.1 THEORETICAL 7
6.2 EXPERIMENT 7

7 LIMIT VALUES 7

7.1 THEORETICAL 7
7.2 EXPERIMENT 8

8 VBIAS & IC 8

8.1 THEORETICAL 8
8.2 EXPERIMENT 8

9 LAB KEY 8

9.1 DC ANALYSIS 8
9.1.1 Simulation results 8
9.2 9.2 FREQUENCY RESPONSE OF AMPLIFIER (GAIN, BW) 9
9.2.1 Simulation results 9
9.2.2 Trial experiment results 9
9.3 POWER CONSUMPTION 9
9.3.1 Simulation results 9
9.4 RIN 9
9.4.1 Simulation results 9
9.5 LIMIT VALUES 9
9.5.1 Simulation results 9
9.5.2 Trial experiment results 9
9.6 V_{BIAS} & I_C 9
9.6.1 Simulation results 9

10 REFERENCES 9

1 PROCEDURE

This experiment was set up to observe the working principle of the [1] PNP Input Active Loaded Amplifier circuit and the effects of changing the parameters in the circuit.

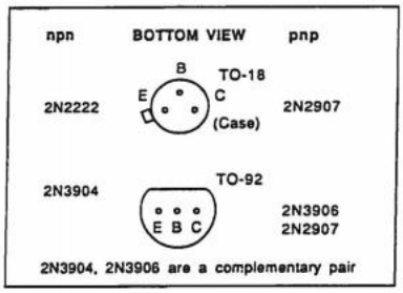


Figure1.2N2222 npn and 2N2907 pnp BJT pin configurations.

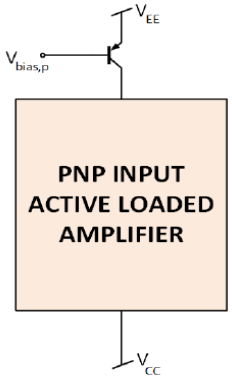


Figure 2. Desired circuit.

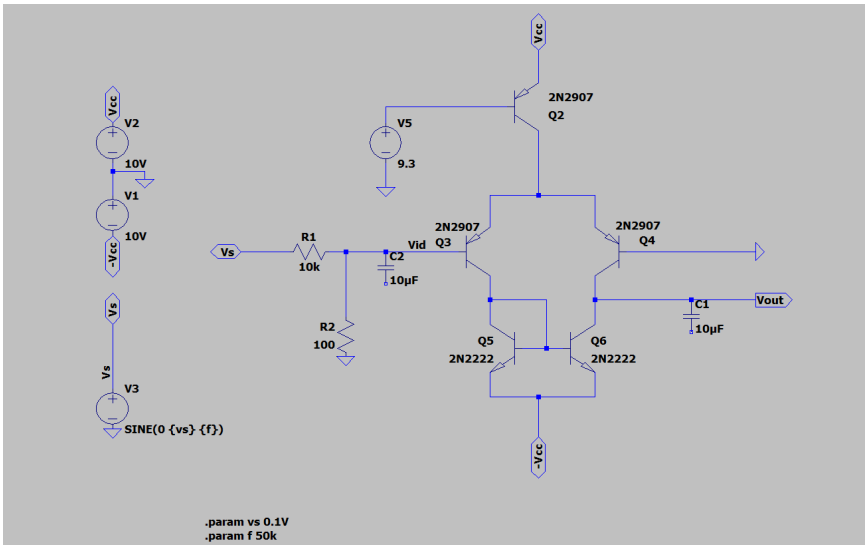


Figure 3. PNP Input Active Loaded Amplifier Circuit in Ltspice.

When the transistors in Figure 1 are used to create the desired circuit in Figure 2, the circuit in Figure 3 is obtained. As shown in Figure 3, 50 kHz, 0.1V (0.2Vpp) signal is connected to the positive input of the differential amplifier and V_{cc}=10 V, -V_{cc}=-10V is selected and the circuit is set up with V_{bias}=9.3. To provide the bias voltage V, a voltage divider was created with a potentiometer of about 10 kΩ. Due to the fact that these elements are not ideal, some results gave different values with simulation. A voltage

divider was applied to the input of the differential amplifier to reduce the input value of 100 mV to 1 mV.

2 DC Analysis

2.1 Theoretical

The inputs were grounded, DC analysis was performed, and the voltage of the Emitter, Base and Collector of the transistors were measured.

Table 1. Voltage value of transistor emitter, base and collector.

V_E(Q2)	V_B(Q2)	V_C(Q2)	V_E(Q3)	V_B(Q3)	V_C(Q3)	V_E(Q4)	V_B(Q4)	V_C(Q4)	V_E(Q5)	V_B(Q5)	V_C(Q5)	V_E(Q6)	V_B(Q6)	V_C(Q6)
10 V	9.3 V	0.68V	0.68V	0 V	-9.3 V	0.68V	0 V	-8.7 V	-10 V	-9.3 V	-9.3 V	-10 V	-9.3 V	-8.7 V

2.2 Experiment

Ground the inputs and measure the Emitter, Base and Collector voltages of the transistors.

Table 2. Voltage value of transistor emitter, base and collector.

V_E(Q2)	V_B(Q2)	V_C(Q2)	V_E(Q3)	V_B(Q3)	V_C(Q3)	V_E(Q4)	V_B(Q4)	V_C(Q4)	V_E(Q5)	V_B(Q5)	V_C(Q5)	V_E(Q6)	V_B(Q6)	V_C(Q6)

3 Frequency Response of Amplifier (Gain, BW)

3.1 Theoretical

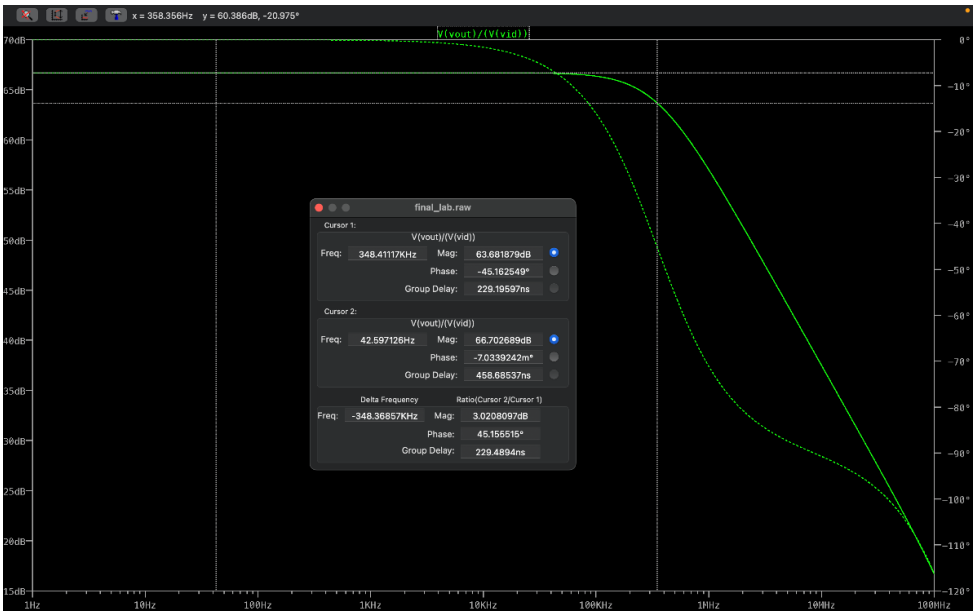


Figure 4. Gain and BW of Circuit.

The gain and BW (the frequency value at which the gain drops by 3dB) of the circuit in Figure 1 are as shown in Figure 4.

3.2 Experiment

Calculate the gain and BW values for V_{bias}=9.3 V by making the voltage measurements in table 3.

Hint: BW is the value at which the gain drops by 3 dB.

Table 3. Measurements to find Gain and BW.

Vs	Vid	Vout	GAIN(Vid/Vout)	BW

4 Theoretical Transient Response of Amplifier (In simulation)

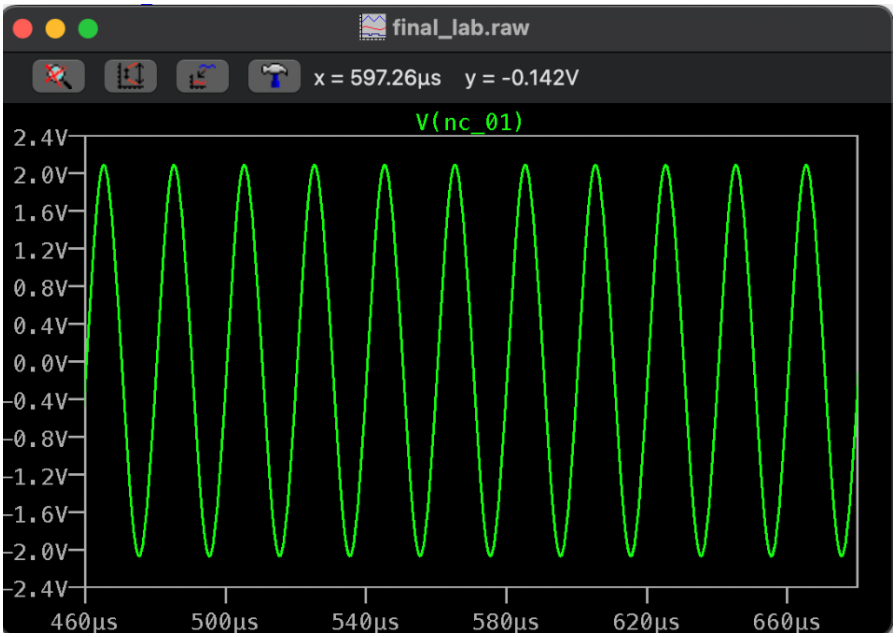


Figure 5. Vout transient AC coupling.

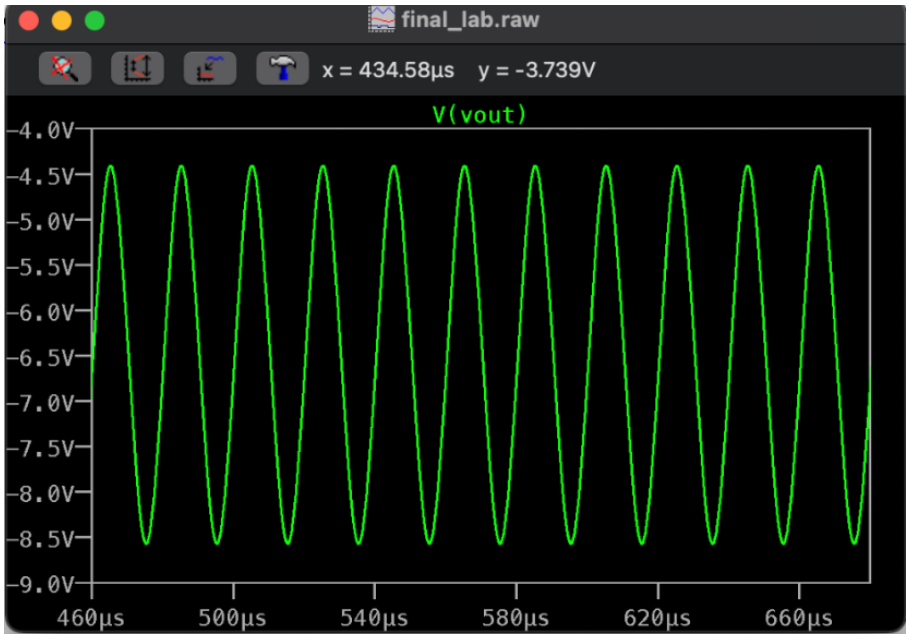


Figure 6. Vout transient DC coupling.

As seen as Figure 5 the output is 4Vpp signal for 0.2Vpp Vs and Vid equal to 2mVpp, so the signal is amplificated with 2000[V/V] gain. From Figure 6, a negative dc offset equal to -6.5 volt is seen it will cause to limit the output voltage from the negative side before the positive side when the amplitude of the input signal increased. The limits will be measured in section “7. Limit values”.

5 Power Consumption

5.1 Theoretical

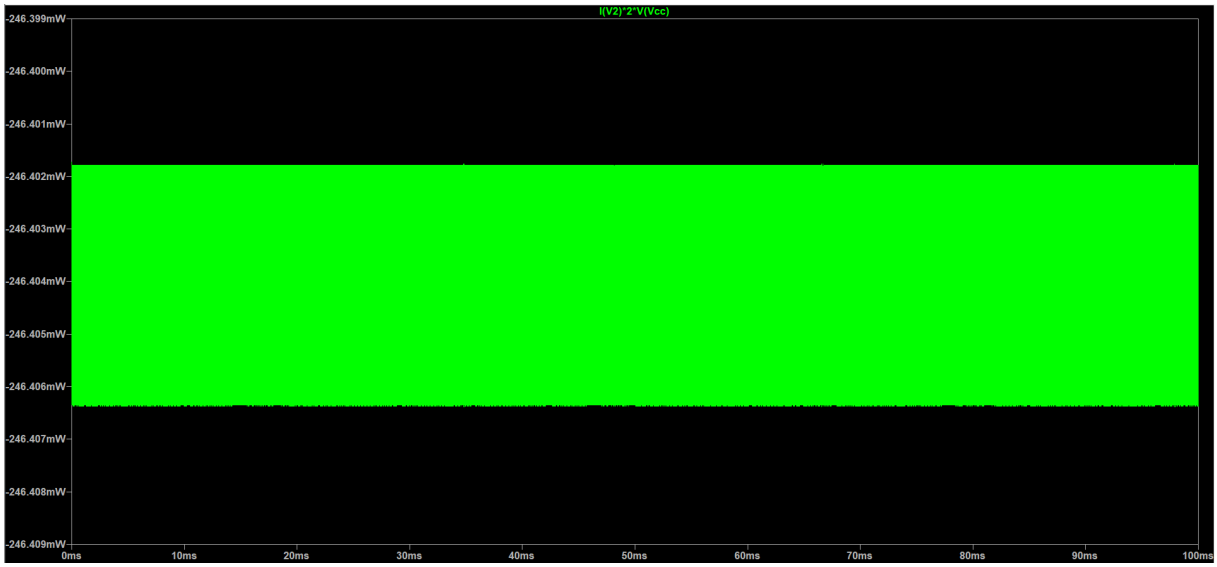


Figure 7. Power consumption of the circuit in Figure 3.

$$P = \big(Vcc - (-Vcc)\big) * Ic$$

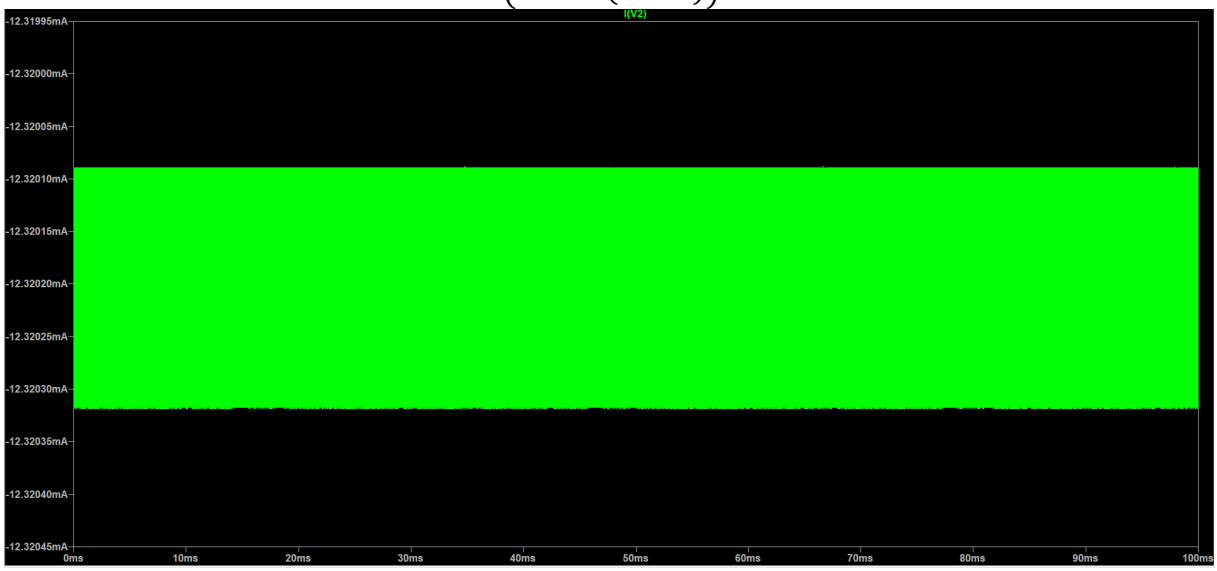


Figure 8. Current flowing from the source.

$$P = \big(10 - (-10)\big) * 12.32\ mA = 246.4\ mW$$

Power consumption in the simulation was verified based on the formula.

5.2 Experiment

Calculate the power consumption of the circuit in Figure 3 in the given area.

Table 4. Power Consumption.

Icc	Power consumption

6 Rin

6.1 Theoretical

$$\frac{Vs - Vb}{Is} = Rin$$

The values in Table 5. are measured and the Rin value is obtained from the above equation.

Table 5. Theoretical Rin.

Vs	Vid	Is	Rin
200 mV	2 mV	20 uA	9900Ω

$$Rin = \frac{200\ mV - 2mV}{20\mu A} = 9900\ \Omega$$

6.2 Experiment

Calculate the value of Rin by measuring the voltage voltages in Table 6.

Table 6. Experiment Rin.

Vs	Vid	Is	Rin

7 Limit Values

7.1 Theoretical

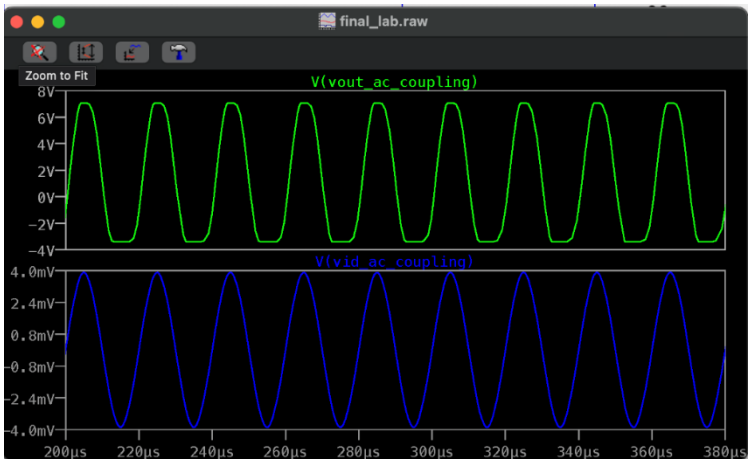


Figure 9. Max limit.

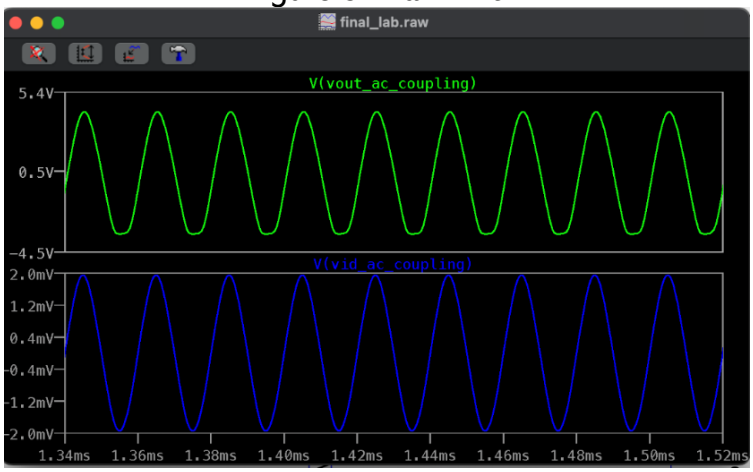


Figure 10. Min Limit.

7.2 Experiment

Observe the upper and lower limits at the Vout output according to the change in Vs value and note them for Vs, Vid and Vout.

Table 7. Limit Values.

Vs+	Vs-	Vid+	Vid-	$Vout^+$	$Vout^-$

8 Vbias & Ic

8.1 Theoretical

When the current (I_c) flowing from the collector (Q2 transistor) is analyzed according to the change in V_{bias}, the graph in Figure 11. appears.

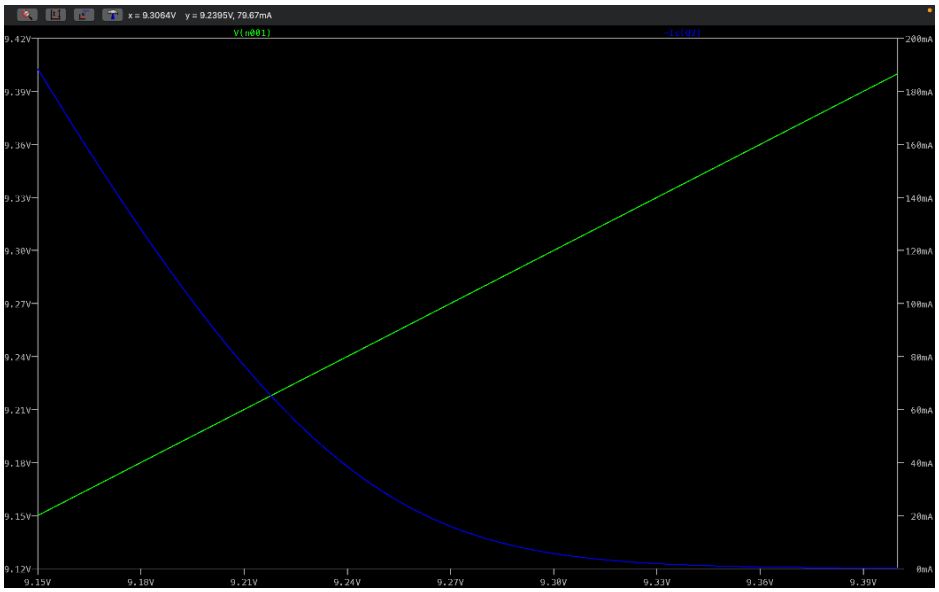


Figure 11. V_{bias} & I_c.

As can be seen in Figure 4, I_c current decreases as V_{bias} increases.

8.2 Experiment

Adjust the V_{bias} values in the circuit according to the table and observe the Ic value.

Table 8. V_{bias} & I_c.

V_{bias}	I_c
9.15 V	
9.30 V	

9 Lab Key

9.1 DC Analysis

9.1.1 Simulation results

Table 9. Voltage value of transistor emitter, base and collector.

V_E(Q2)	V_B(Q2)	V_C(Q2)	V_E(Q3)	V_B(Q3)	V_C(Q3)	V_E(Q4)	V_B(Q4)	V_C(Q4)	V_E(Q5)	V_B(Q5)	V_C(Q5)	V_E(Q6)	V_B(Q6)	V_C(Q6)
10 V	9.3 V	0.68V	0.68V	0 V	-9.3 V	0.68V	0 V	-8.7 V	-10 V	-9.3 V	-9.3 V	-10 V	-9.3 V	-8.7 V

9.2 9.2 Frequency Response of Amplifier (Gain, BW)

9.2.1 Simulation results

Table 10. Gain and BW(peak-tp-peak).

Vs	Vid	Vout	GAIN(Vout/Vid)	BW
200 mV	2 mV	4V	66.70 dB	350 kHz

9.2.2 Trial experiment results

Table 11. Gain and BW(peak-tp-peak)..

Vs	Vid	Vout	GAIN(Vout/Vid)	BW
206 mV	1.98 mV	2.6 V	62.37 dB	210 kHz

9.3 Power Consumption

9.3.1 Simulation results

Table 11. Power consumption.

Icc	Power consumption
12.32 mA	246.4 mW

9.4 Rin

9.4.1 Simulation results

Table 13. Rin.

Vs	Vid	Is	Rin
200 mV	2 mV	20 uA	9900Ω

9.5 Limit Values

9.5.1 Simulation results

Table 14. Limit Values (peak to peak).

Vs+	Vs-	Vid+	Vid-	$Vout^+$	$Vout^-$
0.8 V	0.4 V	8 mV	4 mV	10 V	9V

9.5.2 Trial experiment results

Table 15. Limit Values (peak to peak).

Vs+	Vs-	Vid+	Vid-	$Vout^+$	$Vout^-$
1.2 V	0.85 V	12 mV	8.5mV	10.2 V	9.2 V

9.6 V_{bias} & I_c

9.6.1 Simulation results

Table 16. Vbias & I_c.

V_{bias}	I_c
9.15 V	179.97 mA
9.30 V	11.35 mA

10 References

[1] Smith, K. C. and Sedra, A. S., *Laboratory Explorations for Microelectronic Cirtuits*, Oxford University Press, 1998.